



Change the Shape

Ontario Math C3.2 — read & alter code

Name: _____

Date: _____

★ I can change a loop to make Bit draw a different shape.

Same loop, new shape

Bit's square uses `range(4)` and `right 90`. To draw a different shape, you change TWO things: how many sides (the range number) and the turn (given to you).

Read the chart — each shape just needs a different range and turn.

Bit's square (start here)

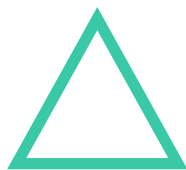
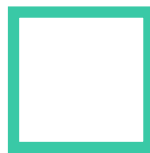
```
for i in range(4):  
    bit.forward(50)  
    bit.right(90)
```

PYTHON

OUTPUT

A square. Change range and the turn to get other shapes.

more sides = bigger range, smaller turn

**triangle**`range(3), right 120`**square**`range(4), right 90`**hexagon**`range(6), right 60`

Change `range()` and the given turn → triangle, square, or hexagon.

1 Read it

A triangle has 3 sides and the turn is 120 (given). What range number do you use? `range(_____)` and `right(_____)`

2 Predict, then run

You change Bit's square to range(6) with right(60). PREDICT: how many sides will the shape have, and what is it called? Then run it.

3 Write your own

Finish the code so Bit draws a TRIANGLE (the turn 120 is given): for i in range(____): bit.forward(50)
bit.right(120)

4 Challenge

Bit's loop is range(5) with right(72). How many sides does the shape have, and what is it called? (Hint: count the repeats.)

Big idea

The range number sets how many sides; the turn is given for each shape. Change those two numbers and you change the shape Bit draws.



Change the Shape

Quick facilitation notes & answer key

What this worksheet teaches

Students take a square's instructions and change two numbers — how many times it repeats and the turn — to draw a different shape (triangle, hexagon, pentagon). The turn for each shape is given. Change the numbers, change the shape. No coding experience needed.

Before you start — showing the drawing (optional)

The worksheet shows the typed instructions AND a picture of each shape, so you can teach straight from the printout — no computer required.

To show it live (optional): open a browser, go to a free "Python turtle" playground (e.g. search "trinket python turtle"), type the few lines from the worksheet, and press Run — a turtle draws it. You only press Run.

How to lead it (about 30 minutes)

1. Start from the square (repeat 4, turn 90). Change it to repeat 3 with turn 120 and run the triangle.
2. Show the turn chart: square 90, triangle 120, pentagon 72, hexagon 60. The repeat number = the number of sides; the turn is given per shape.
3. Let them work. Students do Exercises 1–4, solo or in pairs.
4. Wrap up: the repeat number sets the number of sides; the turn is given for each shape.

Answer key (these are the verified correct answers)

Exercise 1 — repeat 3 times, turn right 120.

Exercise 2 — 6 sides; a hexagon.

Exercise 3 — repeat 3 times.

Exercise 4 (challenge) — 5 sides; a pentagon.

If students get stuck — and ways to adjust

Watch for: changing the repeat number but forgetting the turn must match the shape.

Make it easier: give the turn and have students supply just the repeat number.

Make it harder: ask the repeat number and turn for a hexagon (repeat 6, turn 60).

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, change one, and explain what the change did to the result.

You'll know it worked when

The student matches a shape to its repeat number (Ex 1, 3), predicts the shape after changing the numbers (Ex 2), and names a polygon from the instructions (Ex 4).